

CDF Varname	Item Name (Algorithm Template)	Variable Name in Packet	Starting Bit	padding before	Unsigned nbits
1V_POL_Current	PROCHKADCCHAN01	IDX_TXHDRPROCHKCH0	4	4	12
1.9V_POL_Current	PROCHKADCCHAN01	IDX_TXHDRPROCHKCH0	20	4	12
Proc_Bd_Temp1	PROCHKADCCHAN23	IDX_TXHDRPROCHKCH2	4	4	12
Proc_Bd_Temp2	PROCHKADCCHAN23	IDX_TXHDRPROCHKCH2	20	4	12
1V_Voltage	PROCHKADCCHAN45	IDX_TXHDRPROCHKCH4	4	4	12
FPGA_Temp	PROCHKADCCHAN45	IDX_TXHDRPROCHKCH4	20	4	12
1.9V_Voltage	PROCHKADCCHAN67	IDX_TXHDRPROCHKCH6	4	4	12
3.3V_Voltage	PROCHKADCCHAN67	IDX_TXHDRPROCHKCH6	20	4	12
Detector Voltage	HVPSHKADCCHAN01	IDX_TXHDRHVPSHKCH0	4	4	12
Sensor Voltage	HVPSHKADCCHAN01	IDX_TXHDRHVPSHKCH0	20	4	12
Target Voltage	HVPSHKADCCHAN23	IDX_TXHDRHVPSHKCH2	4	4	12
Reflectron Voltage	HVPSHKADCCHAN23	IDX_TXHDRHVPSHKCH2	20	4	12
Rejection Voltage	HVPSHKADCCHAN45	IDX_TXHDRHVPSHKCH4	4	4	12
Detector Current	HVPSHKADCCHAN45	IDX_TXHDRHVPSHKCH4	20	4	12
Sensor_IP	HVPSHKADCCHAN67	IDX_TXHDRHVPSHKCH6	4	4	12
Sensor_IN	HVPSHKADCCHAN67	IDX_TXHDRHVPSHKCH6	20	4	12
P3.3Ref_HK	LVPSHKADC0CHAN01	IDX_TXHDRLVHK0CH0	4	4	12
P3.3Ref_OP	LVPSHKADC0CHAN01	IDX_TXHDRLVHK0CH0	20	4	12
N6V	LVPSHKADC0CHAN23	IDX_TXHDRLVHK0CH2	4	4	12
P6V	LVPSHKADC0CHAN23	IDX_TXHDRLVHK0CH2	20	4	12
P16V	LVPSHKADC0CHAN45	IDX_TXHDRLVHK0CH4	4	4	12
P3.3V	LVPSHKADC0CHAN45	IDX_TXHDRLVHK0CH4	20	4	12
N5V	LVPSHKADC0CHAN67	IDX_TXHDRLVHK0CH6	4	4	12
P5V	LVPSHKADC0CHAN67	IDX_TXHDRLVHK0CH6	20	4	12
P3.3_IMON	LVPSHKADC1CHAN01	IDX_TXHDRLVHK1CH0	4	4	12
P16V_IMON	LVPSHKADC1CHAN01	IDX_TXHDRLVHK1CH0	20	4	12
P6V_IMON	LVPSHKADC1CHAN23	IDX_TXHDRLVHK1CH2	4	4	12
N6V_IMON	LVPSHKADC1CHAN23	IDX_TXHDRLVHK1CH2	20	4	12
P5V_IMON	LVPSHKADC1CHAN45	IDX_TXHDRLVHK1CH4	4	4	12
N5V_IMON	LVPSHKADC1CHAN45	IDX_TXHDRLVHK1CH4	20	4	12
P2.5V_IMON	LVPSHKADC1CHAN67	IDX_TXHDRLVHK1CH6	4	4	12
N2.5V_IMON	LVPSHKADC1CHAN67	IDX_TXHDRLVHK1CH6	20	4	12
Therm_DECON0	LVPSHKADC2CHAN01	IDX_TXHDRLVHK2CH0	4	4	12
Therm_DECON1	LVPSHKADC2CHAN01	IDX_TXHDRLVHK2CH0	20	4	12
Therm_DAB	LVPSHKADC2CHAN23	IDX_TXHDRLVHK2CH2	4	4	12
Therm_LVPS	LVPSHKADC2CHAN23	IDX_TXHDRLVHK2CH2	20	4	12
Therm_HVPS	LVPSHKADC2CHAN45	IDX_TXHDRLVHK2CH4	4	4	12
P2.5V	LVPSHKADC2CHAN45	IDX_TXHDRLVHK2CH4	20	4	12
N2.5V	LVPSHKADC2CHAN67	IDX_TXHDRLVHK2CH6	4	4	12
Spare**Disregard**	LVPSHKADC2CHAN67	IDX_TXHDRLVHK2CH6	20	4	12

Var_notes
Last measurement in raw DN for Processor Board signal "1V POL Current"
Last measurement in raw DN for Processor Board signal "1.9V POL Current"
Last measurement in raw DN for Processor Board signal "ProcBd Temp1"
Last measurement in raw DN for Processor Board signal "ProcBd Temp2"
Last measurement in raw DN for Processor Board signal "1V Voltage"
Last measurement in raw DN for Processor Board signal "FPGA Temp"
Last measurement in raw DN for Processor Board signal "1.9V Voltage"
Last measurement in raw DN for Processor Board signal "3.3V Voltage"
Last measurement in raw DN for HVPS Board signal "Detector Voltage"
Last measurement in raw DN for HVPS Board signal "Sensor Voltage"
Last measurement in raw DN for HVPS Board signal "Target Voltage"
Last measurement in raw DN for HVPS Board signal "Reflectron Voltage"
Last measurement in raw DN for HVPS Board signal "Rejection Voltage"
Last measurement in raw DN for HVPS Board signal "Detector Current"
Last measurement in raw DN for HVPS Board signal "Sensor IP"
Last measurement in raw DN for HVPS Board signal "Sensor IN"
Last measurement in raw DN for LVPS Board signal "P3.3VREF_HK"
Last measurement in raw DN for LVPS Board signal "P3.3VREF_OP"
Last measurement in raw DN for LVPS Board signal "N6V"
Last measurement in raw DN for LVPS Board signal "P6V"
Last measurement in raw DN for LVPS Board signal "P16V"
Last measurement in raw DN for LVPS Board signal "P3.3V"
Last measurement in raw DN for LVPS Board signal "N5V"
Last measurement in raw DN for LVPS Board signal "P5V"
Last measurement in raw DN for LVPS Board signal "P3.3_IMON"
Last measurement in raw DN for LVPS Board signal "P16V_IMON"
Last measurement in raw DN for LVPS Board signal "P6V_IMON"
Last measurement in raw DN for LVPS Board signal "N6V_IMON"
Last measurement in raw DN for LVPS Board signal "P5V_IMON"
Last measurement in raw DN for LVPS Board signal "N5V_IMON"
Last measurement in raw DN for LVPS Board signal "P2.5V_IMON"
Last measurement in raw DN for LVPS Board signal "N2.5V_IMON"
Last measurement in raw DN for LVPS Board signal "THERM_DECON0"
Last measurement in raw DN for LVPS Board signal "THERM_DECON1"
Last measurement in raw DN for LVPS Board signal "THERM_DAB"
Last measurement in raw DN for LVPS Board signal "THERM_LVPS"
Last measurement in raw DN for LVPS Board signal "THERM_HVPS"
Last measurement in raw DN for LVPS Board signal "P2.5V"
Last measurement in raw DN for LVPS Board signal "N2.5V"
Last measurement in raw DN for LVPS Board signal "Spare"

Plain Language Summary	Units
High side current reading on 1.0V bus on Processor Board (ADC channel 0)	mA
High side current reading on 1.9V bus on Processor Board (ADC channel 1)	mA
Temperature reading near high speed ADCs on Processor Board (ADC channel 2)	C
Temperature reading near center of Processor Board (ADC channel 3)	C
Voltage reading of 1.0V bus on Processor Board (ADC channel 4)	V
Temperature reading on FPGA on Processor Board (ADC channel 5)	C
Voltage reading of 1.9V bus on Processor Board (ADC channel 6)	V
Voltage reading of 3.3V bus on Processor Board (ADC channel 7)	V
Voltage reading of Detector on HVPS Board (ADC Channel 0)	V
Voltage reading of Sensor on HVPS Board (ADC Channel 1)	V
Voltage reading of Target on HVPS Board (ADC Channel 2)	V
Voltage reading of Reflectron on HVPS Board (ADC Channel 3)	V
Voltage reading of Rejection on HVPS Board (ADC Channel 4)	V
Reading of Detector current on HVPS Board (ADC Channel 5)	mA
Reading of HVPS Sensor positive current (ADC Channel 6)	mA
Reading of HVPS Sensor negative current (ADC Channel 7)	mA
Positive voltage reading on +3.3V bus Housekeeping reference on LVPS Board (ADC 0	V
Positive voltage reading on +3.3V bus Operations reference on LVPS Board (ADC 0 channel	V
Negative voltage reading on -6V bus on LVPS Board (ADC 0 channel 2)	V
Positive voltage reading on +6V bus on LVPS Board (ADC 0 channel 3)	V
Positive voltage reading on +16V bus on LVPS Board (ADC 0 channel 4)	V
Positive voltage reading on +3.3V bus on LVPS Board (ADC 0 channel 5)	V
Negative voltage reading on -5V bus on LVPS Board (ADC 0 channel 6)	V
Positive voltage reading on +5V bus on LVPS Board (ADC 0 channel 7)	V
Positive current reading on +3.3V bus on LVPS Board (ADC 1 channel 0)	A
Positive current reading on +16V bus on LVPS Board (ADC 1 channel 1)	A
Positive current reading on +6V bus on LVPS Board (ADC 1 channel 2)	A
Negative current reading on -6V bus on LVPS Board (ADC 1 channel 3)	A
Positive current reading on +5V bus on LVPS Board (ADC 1 channel 4)	A
Negative current reading on -5V bus on LVPS Board (ADC 1 channel 5)	A
Positive current reading on +2.5V bus on LVPS Board (ADC 1 channel 6)	A
Negative current reading on -2.5V bus on LVPS Board (ADC 1 channel 7)	A
Temperature reading of Target (Primary), used for decon control loop (ADC 2 channel 0)	C
Temperature reading of Target (Redundant), used for decon control loop (ADC 2 channel 1)	C
Temperature reading on Decontamination and Actuation Board (DAB) (ADC 2 channel 2)	C
Temperature reading on LVPS Board (ADC 2 channel 3)	C
Temperature reading on HVPS Board (ADC 2 channel 4)	C
Positive voltage reading on +2.5V bus on LVPS Board (ADC 2 channel 5)	V
Negative voltage reading on -2.5V bus on LVPS Board (ADC 2 channel 6)	V
Spare input on LVPS Board (ADC 2 channel 7)	N/A

DN Range	CO	C1	C2	C3	C4	C5
0 - 4095	0	0.9029253	0	0	0	0
0 - 4095	0	0.9029253	0	0	0	0
0 - 2063	162.8276	-0.2668687	0.000332379	-2.43E-07	9.03E-11	-1.33E-14
2064 - 3853	714.2124	-1.194011	0.000859909	-3.15E-07	5.75E-11	-4.21E-15
3854 - 4051	136881.2	-105.4644	0.02710281	-2.32E-06	0	0
4052 - 4095	-100809.9	50.03809	-0.00621209	0.00E+00	0	0
0 - 2063	162.8276	-0.2668687	0.000332379	-2.43E-07	9.03E-11	-1.33E-14
2064 - 3853	714.2124	-1.194011	0.000859909	-3.15E-07	5.75E-11	-4.21E-15
3854 - 4051	136881.2	-105.4644	0.02710281	-2.32E-06	0	0
4052 - 4095	-100809.9	50.03809	-0.00621209	0.00E+00	0	0
0 - 4095	0	0.000805861	0	0	0	0
0 - 4095	358.31	-0.1187104	0	0	0	0
0 - 4095	0	0.000805861	0	0	0	0
0 - 4095	0	0.001611722	0	0	0	0
0 - 4095	0	1.4652	0	0	0	0
0 - 4095	0	1.4652	0	0	0	0
0 - 4095	0	1.4652	0	0	0	0
0 - 4095	0	1.4652	0	0	0	0
0 - 4095	0	1.4652	0	0	0	0
0 - 4095	0	7.33E-06	0	0	0	0
0 - 4095	0	2.43E-05	0	0	0	0
0 - 4095	0	2.43E-05	0	0	0	0
0 - 4095	0	0.00161172	0	0	0	0
0 - 4095	0	0.00161172	0	0	0	0
0 - 4095	-33	0.00886447	0	0	0	0
0 - 4095	0	0.00241758	0	0	0	0
0 - 4095	0	0.00482711	0	0	0	0
0 - 4095	0	0.00161172	0	0	0	0
0 - 4095	-33	0.00886447	0	0	0	0
0 - 4095	0	0.00161172	0	0	0	0
0 - 4095	0	0.00161172	0	0	0	0
0 - 4095	0	0.000161172	0	0	0	0
0 - 4095	0	8.06E-05	0	0	0	0
0 - 4095	0	-8.06E-05	0	0	0	0
0 - 4095	0	4.03E-05	0	0	0	0
0 - 4095	0	-2.01E-05	0	0	0	0
0 - 4095	0	0.00161172	0	0	0	0
0 - 4095	0	-4.03E-05	0	0	0	0
0 - 4095	-110.293	0.0665351	0.000305738	-6.32E-07	5.99E-10	-2.99E-13
0 - 4095	-150.478	0.376986	-0.00062072	7.74E-07	-5.82E-10	2.56E-13
0 - 4095	-443.248	1.31224	-0.00180422	1.40E-06	-6.44E-10	1.74E-13
0 - 2091	200.4137	-0.2020699	0.000175403	-9.74E-08	2.89E-11	-3.55E-15
2091 - 3869	1193.93	-1.97726	0.0014106	-5.08E-07	9.12E-11	-6.55E-15
3869 - 4051	170704.7	-131.1981	0.03363742	-2.88E-06	0.00E+00	0.00E+00
4051 - 4095	-104713.9	52.02446	-0.006463374	0.00E+00	0.00E+00	0.00E+00
0 - 2091	200.4137	-0.2020699	0.000175403	-9.74E-08	2.89E-11	-3.55E-15
2091 - 3869	1193.93	-1.97726	0.0014106	-5.08E-07	9.12E-11	-6.55E-15
3869 - 4051	170704.7	-131.1981	0.03363742	-2.88E-06	0.00E+00	0.00E+00
4051 - 4095	-104713.9	52.02446	-0.006463374	0.00E+00	0.00E+00	0.00E+00
0 - 4095	0	0.00161172	0	0.00E+00	0.00E+00	0.00E+00
0 - 4095	-33	0.00886447	0	0	0	0
0 - 4095	0	0.00161172	0	0	0	0

